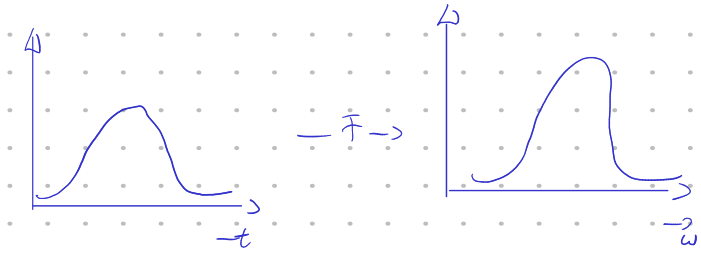
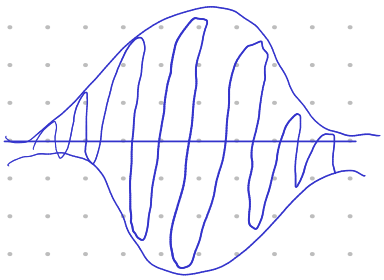
AOFC 7 04.12.2025

Introduction:

many phenomena can be expressed as gaussian pulses.

If the gaussian is transform limited the shape in frequency and time domain is the same





→ we now consider a chirped pulse, a derivation of the carrier frequency inside our pulse.

up chirp → frequency rises in pulse duration

down- chirp → frequency lowers . —"—

expression for a gaussian pulse:

$$A(0,t) = A_0 \cdot \exp\left(-\frac{1+ic}{2}\left(\frac{t}{T_0}\right)^2\right)$$

A_0: peak Amplitude

c: chirp parameter

T_0: half width of the pulse

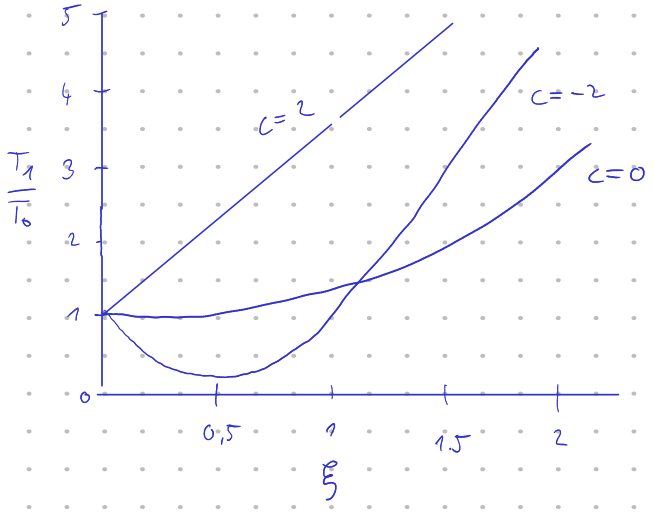
$$T_{FWHM} = T_0 \cdot 1.665$$

a Gaussian pulse broadens or shrinks in time, if the medium is said to be dispersive,

How big is this broadening in time?

$$\frac{T_1}{T_0} = \left[1 + \left(\frac{C \beta_2 \cdot z}{T_0^2} \right)^2 + \left(\frac{\beta_2 \cdot z}{T_0^2} \right)^2 \right]^{1/2}$$

$L_D = T_0^2 / |\beta_2|$ dispersion length. $\xi = z / L_D$



for $C\beta_2 > 0$ pulse broadens monotonically

$C\beta_2 < 0$ pulse shrinks initially, will be minimum at:

$$z_{min} = \frac{|C|}{1 + C^2} L_D$$

$$T_1^{min} = T_0 / (1 + C^2)^{1/2}$$

limit for the bitrate:

$$\sigma = \left[\langle t \rangle^2 - \langle t^2 \rangle \right]^{1/2}$$

this is complicated to calculate,

only remember $\sigma_0 = \frac{T_0}{\sqrt{2}}$ $\sigma_0 = \frac{FWHM}{2.355}$

now, how much is the pulse broadend after a fiber?

$$\sigma_{tot} = \left(\sigma_0^2 + \sigma_D^2 \right)^{1/2}$$

for what is this useful?

we now can start to define a dispersion limit for the bitrate:

$T_B = 1/B$ $B \hat{=}$ bit rate.

$$\boxed{\sigma_{tot} \leq T_B / 4}$$

Exercise 12

a) $\tau_{FWHM} = 10\,ns$ $\quad \lambda_{source} = 30\,nm$ (FWHM) $\quad L = 10\,km$ $\quad D = -80\,ps/(nm\,km)$

$$\sigma_0 = \frac{10\,ns}{2.355} = 4.25\,ns \qquad \sigma_\lambda = \frac{30\,nm}{2.355} = 12.74\,nm$$

$$\sigma_D = |D| \cdot L \cdot \sigma_\lambda = \left|-80\,\frac{ps}{nm\,km}\right| \cdot 10\,km \cdot 12.74\,nm = 10.19\,ns$$

$$\sigma_{tot} = \sqrt{\left(10.19\,ns^2 + 4.25\,ns^2\right)} = 11.04\,ns$$

$$B_{max} = \frac{0.25}{11.04\,ns} = 22.6\,Mbit/s \rightarrow \text{only for on/off keying (one bit per symbol)}$$

b) $L = 60\,km$ $\quad \lambda_1 = 1.3\,\mu m$ $\quad \beta_2 = 0$

$\lambda_2 = 1.55\,\mu m$ $\quad \beta_2 = -20\,\frac{ps^2}{km}$

$$D = -\frac{2\pi c}{\lambda^2}\beta_2$$

$D_1 = 0$

$$D_2 = -\frac{2\pi \cdot 3e8\,\frac{m}{s}}{1550\,nm}\left(-20\,\frac{ps^2}{km}\right) = 15.7\ ps/(nm\,km)$$

$\sigma_{D_2} = |D| \cdot L \cdot \sigma_\lambda$ ← where is σ_λ?

pulse is transform limited!

the spectral width of Fourier transform (time domain), equals the optical spectrum.

$$\sigma_\omega \cdot \sigma_0 = 1/2 \qquad \sigma_\lambda = \frac{\lambda^2}{2\pi \cdot c_0}\cdot\sigma_\omega = \frac{\lambda^2}{4\pi c \cdot \sigma_0}$$

$$\sigma_{\lambda_2} = \frac{1550^2}{4\pi c \cdot \frac{50\,ps}{2.335}} = 0.03\,nm! \rightarrow \text{bandwidth of the pulse!}$$

$$\sigma_{D_2} = 60\,km \cdot \left|17.5\,\frac{ps}{nm\,km}\right| \cdot 0.03\,nm = 28.3\,ps$$

$$\sigma_{tot_2} = \left(28.3\,ps^2 + \left(\frac{50\,ps}{2.335}\right)^2\right)^{1/2} = 35,4\,ps \rightsquigarrow \frac{0.25}{35,4\,ps} = \boxed{7,1\,Gbit}$$

@ $\lambda = 1.3\ \mu m$

$$\sigma_{tot} = \frac{50\ ps}{2.335} = 21.2\ ps \qquad \frac{0.25}{21.2\ ps} = \boxed{11.8\ Gbit}$$

c) $\lambda = 1.55\ \mu m$, $5\ Gbit/s$, Gaussian @ 100ps FWHM, chirp=-6

$$\sigma_0 = \frac{100\ ps}{2.335} = 42.2\ ps \quad , \quad T_0 = \frac{100}{1.665} = 60\ ps$$

Maximum RMS width $\sigma_{MAX} = \frac{0.25}{B} = \frac{0.25}{5\ Gbit/s} = 50\ ps$

$$\left(\frac{\sigma_{max}}{\sigma_0}\right)^2 = \left(\frac{50}{42.4}\right)^2 = 1.39 \rightarrow \text{Maximum allowed broadening factor for } 5\ \frac{Gbit}{s}$$

$$\left(\frac{\sigma_{MAX}}{\sigma_0}\right)^2 = \left(1 + \frac{C\,\beta_2\,L}{T_0^2}\right)^2 + \left(\frac{\beta_2\,L}{T_0^2}\right)^2$$

now $\frac{\beta_2\,L}{T_0^2} = \frac{-20 \cdot L}{(60ps)^2} = \frac{-20\,L}{3600} = \frac{-L}{180} = S$

$$\left(\frac{\sigma_{MAX}}{\sigma_0}\right)^2 = (1 + C \cdot S)^2 + S^2$$

$$1.39 = \left(1 + 6 \cdot \frac{L}{180}\right)^2 + \frac{L^2}{180^2}$$

$$1.39 = 1 + \frac{2L}{36} + \frac{L^2}{90} + \frac{L^2}{32400}$$

$$0 = 37L^2 + 2160L - 12636$$

$$L = \frac{-2160 + \sqrt{(2160^2 + 4 \cdot 37 \cdot 12636)}}{74}$$

$$L = 5{,}2\ km$$

unchirped pulses

$$\left(\frac{\sigma_{MAX}}{\sigma_0}\right)^2 \approx 1 + S^2 \;!$$

$$1.39 \approx 1 + \frac{L^2}{32400}$$

$$\sqrt{0.39 \cdot 32400} = L$$

$$L = 112\,km\,!$$